

# Empirical Proof Record: VALIDATED

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## Empirical Proof Record — VALIDATED

**Proof ID:** 0168ed60ac7dd04a050e4a1d7eea32be...

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### 1. Claim

Across 30 synthetic (x, y) pairs of size 100 (seed=20260518), “`scipy.stats.pearsonr`“, “`numpy.corrcoef`“, and “`pingouin.corr(method='pearson')`“ produce Pearson r values that agree to  $\leq 1e-09$  across all three pairwise comparisons. (RFC 0002 Phase E1: cross-framework validation at the statistical-primitive layer; three-way variant.)

- **Operationalisation:** Generate N (x, y) pairs where  $x \sim \text{Normal}(0, 1)$  and  $y = \rho_{\text{target}} \times x + \sqrt{1 - \rho_{\text{target}}^2} \times \text{Normal}(0, 1)$  for a sweep of  $\rho_{\text{target}}$  across  $[-0.9, 0.9]$ . Compute Pearson r under each of `scipy`, `numpy`, `pingouin`. Compute `max_abs_diff` = max over (pair, backend-comparison) of  $|r_a - r_b|$ . VALIDATED iff `max_abs_diff`  $\leq$  tolerance across all three pairwise comparisons.
- **Threshold:** `max_absolute_pearson_difference`  $\leq 1e-09$  correlation
- **H0:** At least one backend pair (`scipy` $\leftrightarrow$ `numpy`, `scipy` $\leftrightarrow$ `pingouin`, `numpy` $\leftrightarrow$ `pingouin`) disagrees on Pearson r by more than the floating-point tolerance.
- **H1:** All three backends compute identical Pearson r to floating-point tolerance across every pair.

### 2. Verdict

**VALIDATED** — observed 3.3306690738754696e-16

30 pairs  $\times$  3 backends; max pairwise  $|\Delta r| = 3.331e-16$  on `scipy_vs_numpy` ( $\leq 1e-09$  tol)

### 3. Evidence

| Pillar                             | Statistic                       | Value                 |
|------------------------------------|---------------------------------|-----------------------|
| pearson_scipy_vs_numpy_vs_pingouin | max_absolute_pearson_difference | 3.3306690738754696e-1 |

### 4. Pre-registration

- Registered at: 2026-05-31T18:33:55.861596+00:00
- Config hash: 395ced3ae94d681473ffeddcaa298fd2...
- Data hash: 395ced3ae94d681473ffeddcaa298fd2...

### 5. Signature

20b2ae6eb7090a12bf3d2ea7621d5b17300898c5b4b71addfa98baa34b58dc14